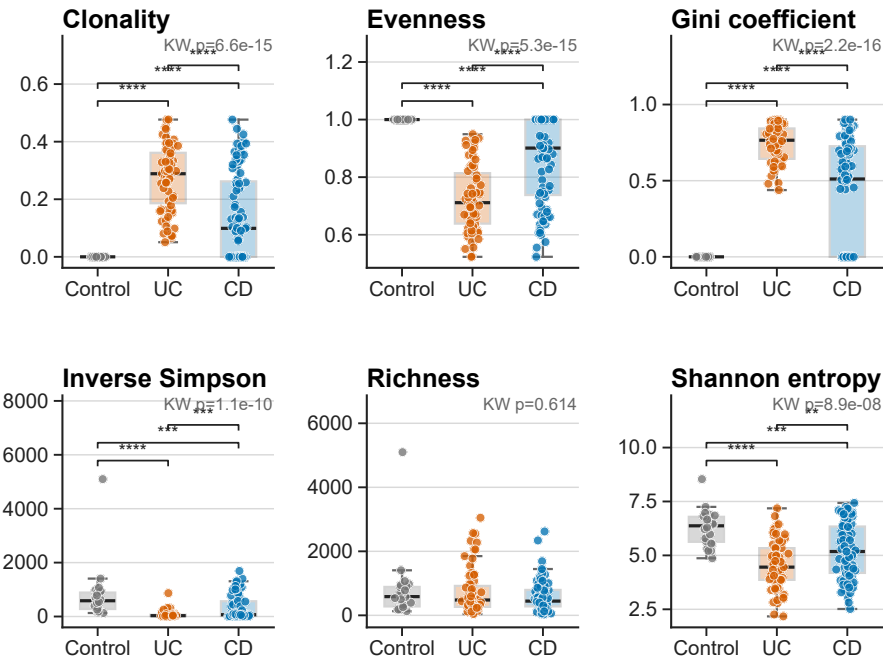
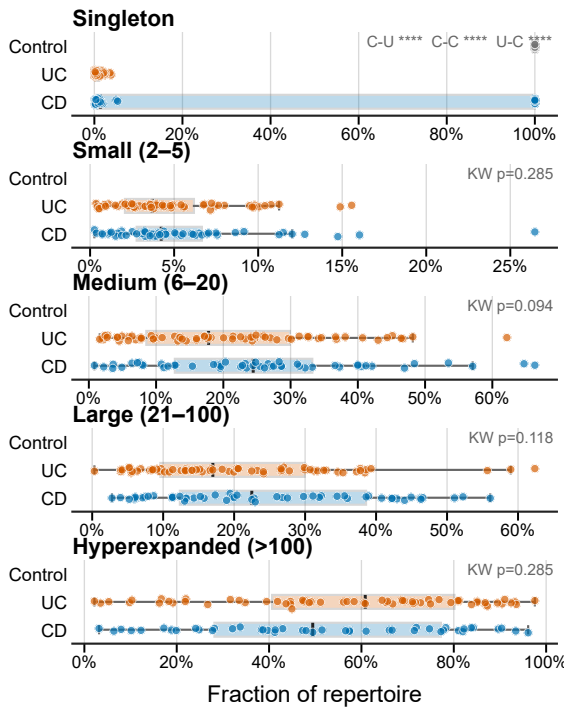


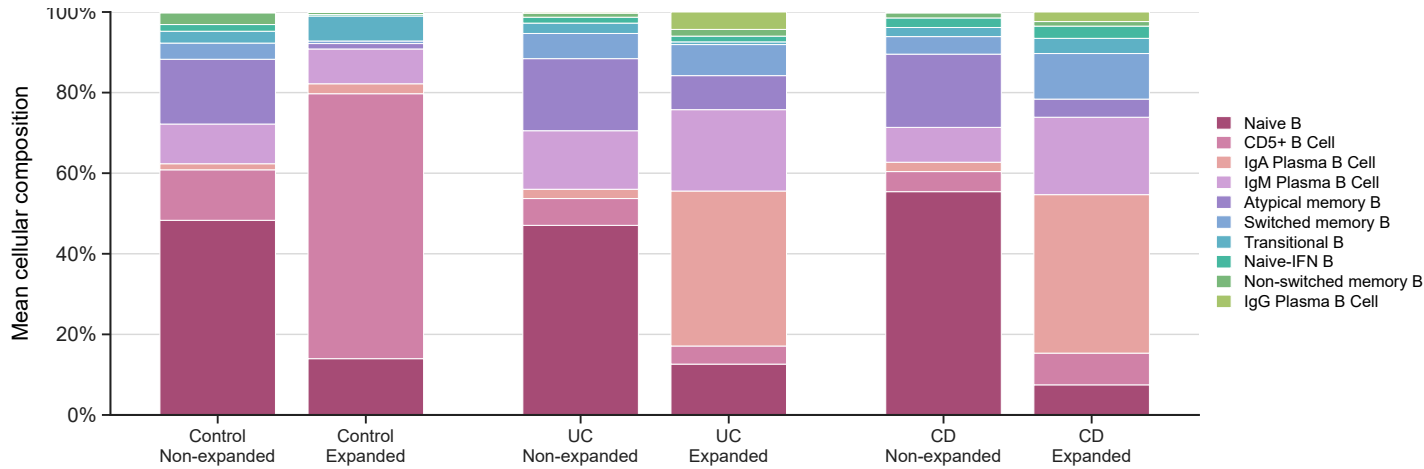
A Participant-level repertoire diversity



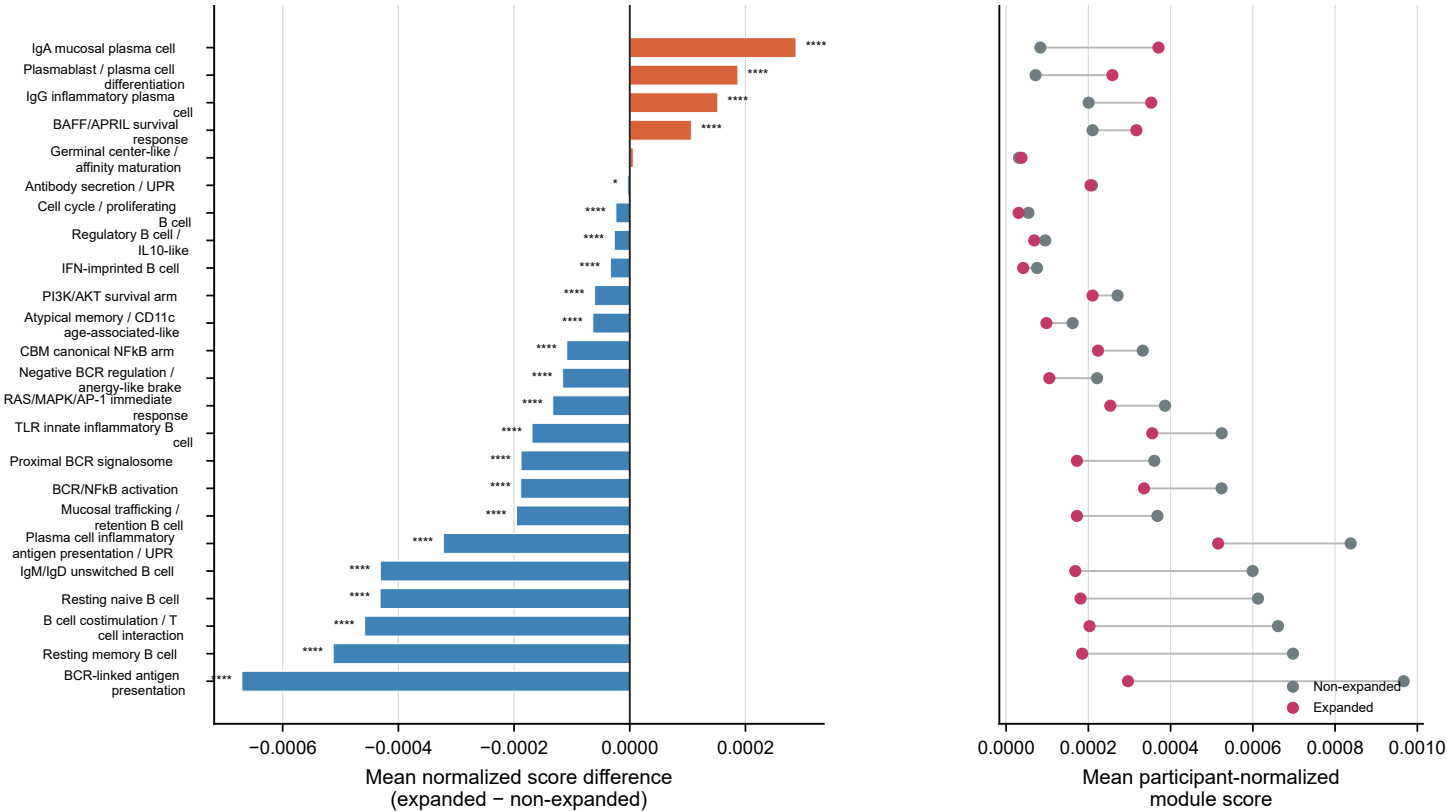
B Clone-size architecture



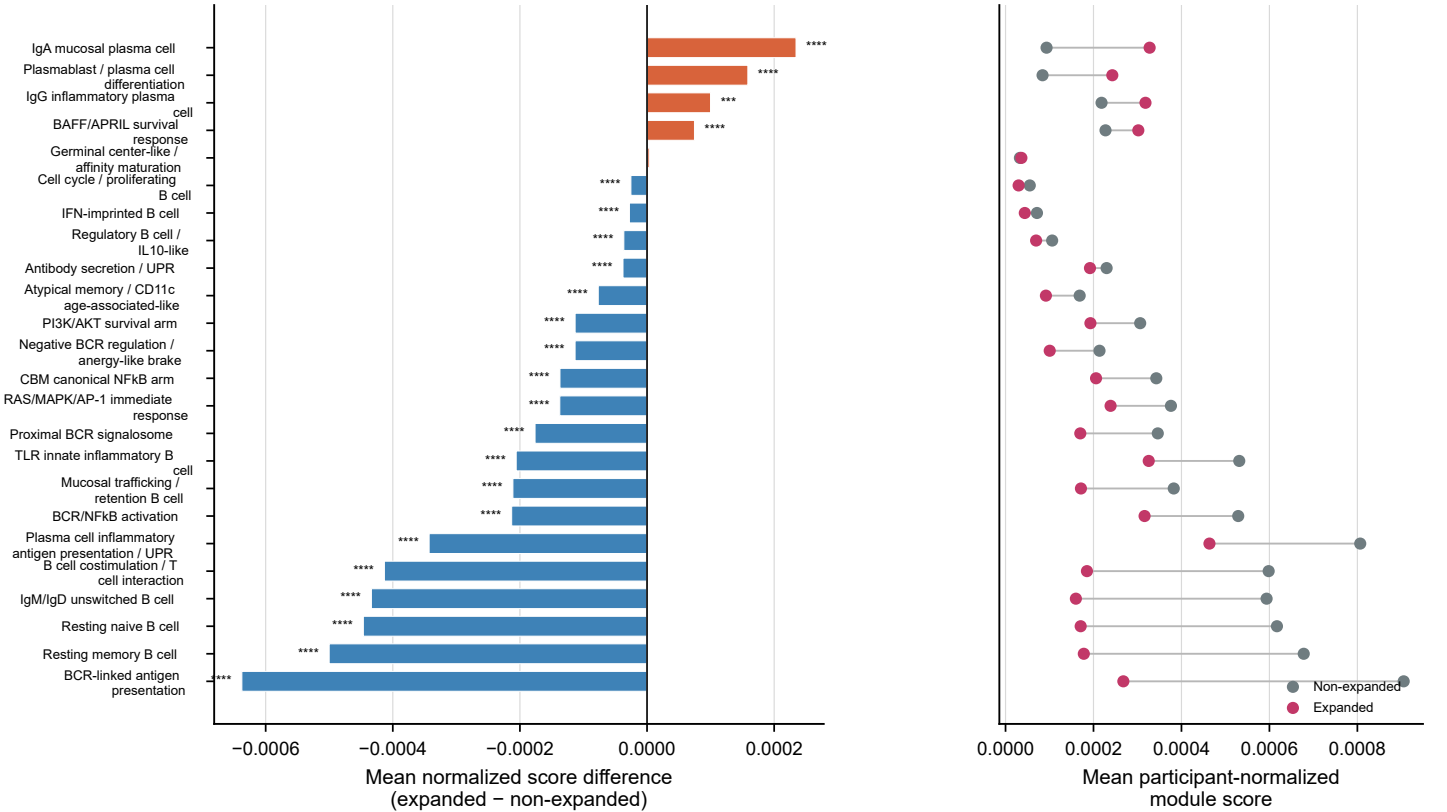
C B-cell-state composition of exact paired heavy-light clonotypes



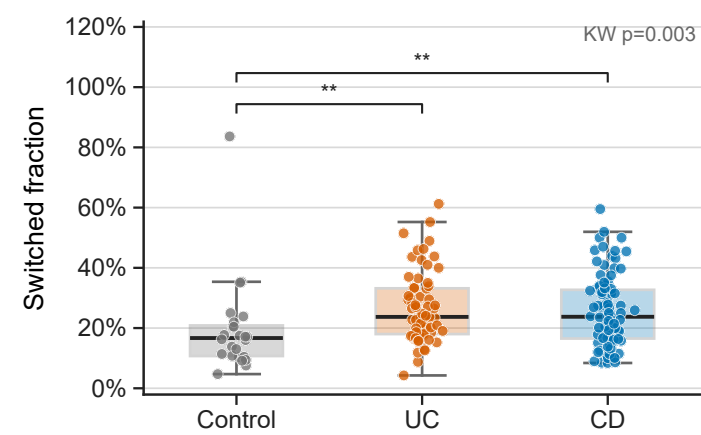
D Crohn's disease: expansion-linked B-cell programs



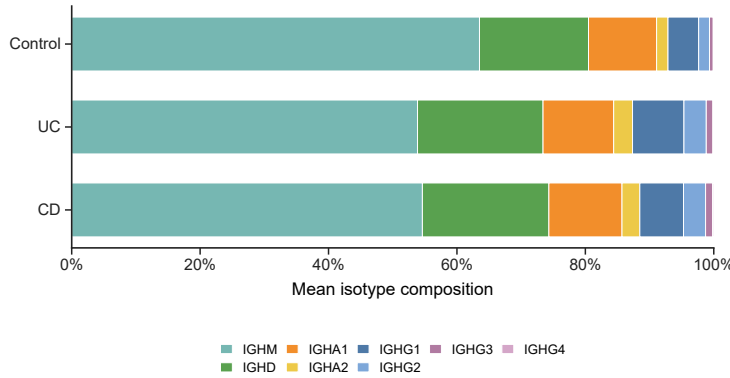
E Ulcerative colitis: expansion-linked B-cell programs



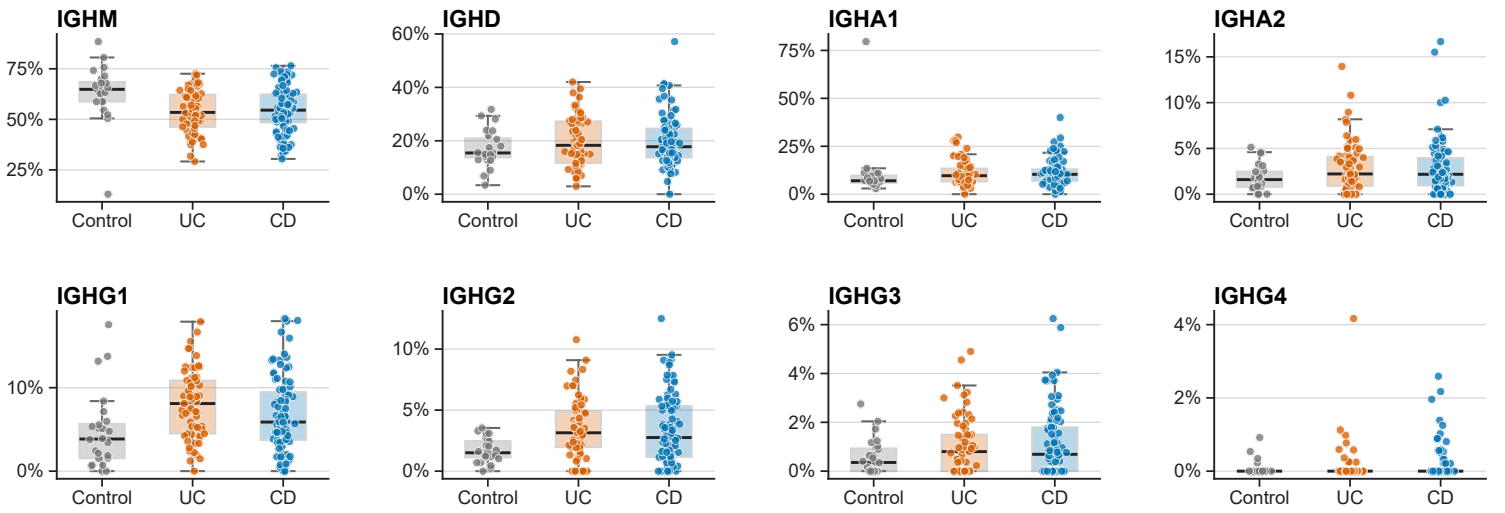
F Class-switched B cells



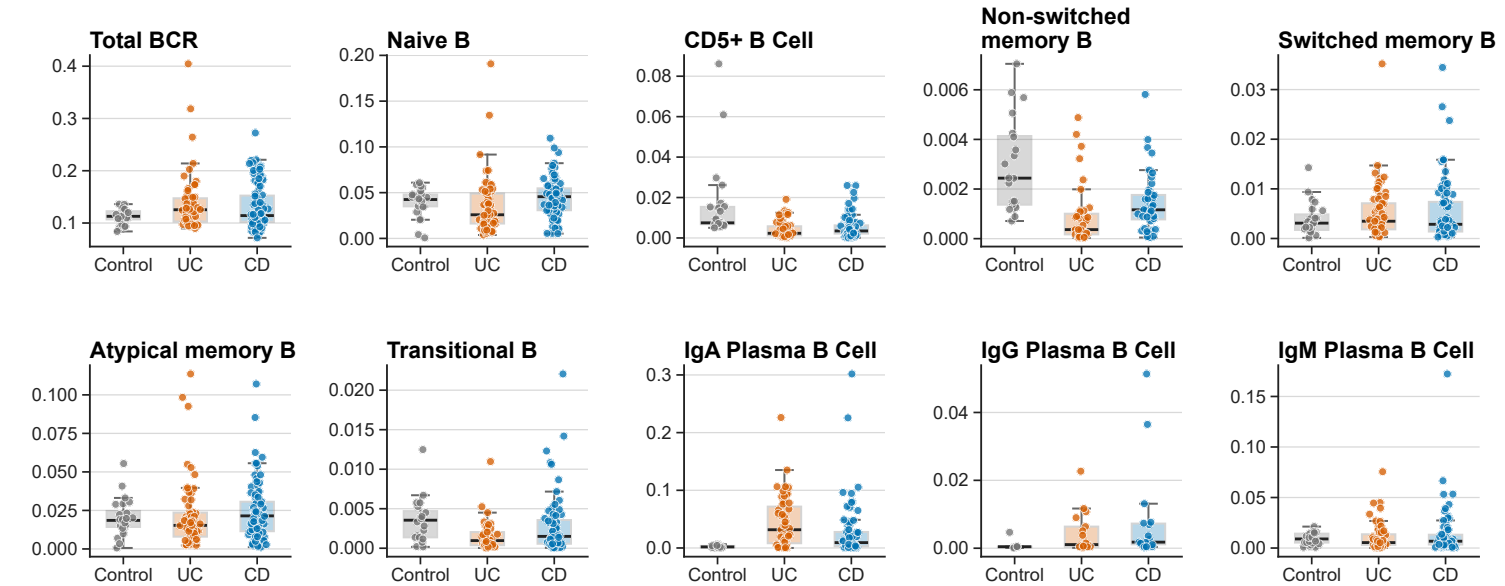
G Heavy-chain isotype composition



Participant-level isotype fractions (G)



H Somatic hypermutation across B-cell states



Boxes show median and interquartile range; points denote participants. *FDR < 0.05; **FDR < 0.01; ***FDR < 0.001; ****FDR < 0.0001.